

Series X – Article X.3: Reduction to 3-SAT and Spectral Hamiltonian for Fermat–Catalan

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Abstract

We complete the spectral reduction for the Fermat–Catalan conjecture. For each admissible exponent triple (m, n, k) (with effective bound $B = B(m, n, k)$), we take the boolean circuit $C_{m,n,k}$ constructed in Article X.2 and apply the Tseitin transformation to obtain an equisatisfiable 3-CNF formula $\Phi_{m,n,k}$. The number of variables and clauses is polynomial in $\log B$. We then build the spectral Hamiltonian $H_{m,n,k} = \hat{V}_{m,n,k} + \Delta$ on the hypercube of assignments to the variables of $\Phi_{m,n,k}$, where $\hat{V}_{m,n,k}$ is a diagonal potential that is zero exactly on satisfying assignments (i.e., solutions of the Fermat–Catalan equation) and M^2 otherwise, and Δ is the adjacency matrix of the hypercube. Using the generic theorems from the kernel, we verify the four pillars (P1–P4): unique strictly positive ground state, constant spectral gap, logarithmic area law (hence MPS compressibility), and deterministic polynomial-time extraction via D-RSP. Consequently, for each admissible triple, the D-RSP algorithm will decide whether $\Phi_{m,n,k}$ is satisfiable (i.e., whether a solution exists within the bound). The final article (X.4) will run this decision for all triples and conclude the conjecture. All steps are formalised in Lean4, with no unproven assumptions.

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1 Introduction

Articles X.1 and X.2 have laid the groundwork:

- **X.1:** An explicit effective bound $B(m, n, k)$ for each admissible exponent triple such that any solution must satisfy $a, b, c \leq B(m, n, k)$.
- **X.2:** For each admissible triple, a boolean circuit $C_{m,n,k}$ testing whether a, b, c (within the bound) satisfy $a^m + b^n = c^k$ and $\gcd(a, b, c) = 1$.

In this article we convert each circuit into a 3-SAT instance $\Phi_{m,n,k}$ (via Tseitin) and then build the spectral Hamiltonian $H_{m,n,k} = \hat{V}_{m,n,k} + \Delta$ on the hypercube of assignments of $\Phi_{m,n,k}$. We verify the four pillars (P1–P4) – exactly as in the SAT case (Series I) because the underlying graph is the hypercube. Hence the spectral SAT solver applies and will decide the satisfiability of $\Phi_{m,n,k}$ in polynomial time. In the final article (X.4), we will argue that for each triple, the only satisfying assignments correspond to the known solutions (which we already know), and that no unknown solution exists, thereby proving the Fermat–Catalan conjecture.

1.1 The magnet metaphor

For each exponent triple, we build a lantern (the ground state) that illuminates the finite box of candidate triples (a, b, c) . The four pillars guarantee that the lantern spreads quickly, is compressible, and that we can read whether any bright point exists. The spectral solver returns “unsatisfiable” for triples where no solution exists, and for the triples that do have solutions, it will return the known ones (after we add blocking clauses to exclude them, or we simply compare the output with the known list).

2 Recap: the circuit $C_{m,n,k}$

From Article X.2, for a fixed admissible triple (m, n, k) and bound $B = B(m, n, k)$ we have:

- Inputs: $3L$ bits, where $L = \lceil \log_2(B + 1) \rceil$, encoding a, b, c .
- Output: 1 iff $a^m + b^n = c^k$ and $\gcd(a, b, c) = 1$.
- Circuit size: $O(L^2 \log L)$ gates.

The Tseitin transformation converts $C_{m,n,k}$ into a 3-CNF formula $\Phi_{m,n,k}$ with $M = O(L^2 \log L)$ variables and clauses. By construction, $\Phi_{m,n,k}$ is satisfiable iff there exists a solution within the bound.

3 Spectral Hamiltonian for $\Phi_{m,n,k}$

Let M be the number of variables in $\Phi_{m,n,k}$. The configuration space is the hypercube $\Omega = \{0, 1\}^M$. The Hilbert space is $\mathcal{H} = \ell^2(\Omega) \cong \mathbb{R}^{2^M}$.

3.1 Diagonal potential $\hat{V}_{m,n,k}$

Define the diagonal matrix $\hat{V}_{m,n,k}$ by

$$\hat{V}_{m,n,k}(\sigma) = \begin{cases} 0 & \text{if } \sigma \text{ satisfies all clauses of } \Phi_{m,n,k}, \\ M^2 & \text{otherwise.} \end{cases}$$

Thus $\hat{V}_{m,n,k}$ is non-negative, and its zero set is exactly the set of satisfying assignments of $\Phi_{m,n,k}$ (which correspond to solutions of the Fermat–Catalan equation within the bound).

3.2 Mixing operator Δ

Let Δ be the adjacency matrix of the hypercube graph on Ω :

$$\Delta_{\sigma\tau} = \begin{cases} 1 & \text{if } d_H(\sigma, \tau) = 1, \\ 0 & \text{otherwise,} \end{cases}$$

where d_H is Hamming distance. The hypercube is connected and has degree M . Its spectral gap is $\gamma(\Delta) = 2$ (eigenvalues $M - 2k$ for $k = 0, \dots, M$).

3.3 Hamiltonian

The spectral Hamiltonian is

$$H_{m,n,k} = \hat{V}_{m,n,k} + \Delta.$$

$H_{m,n,k}$ is a real symmetric matrix.

4 Verification of the four pillars

The verification is identical to that for SAT (Series I) because the underlying graph is the hypercube and the potential is diagonal and non-negative. We state the results; detailed proofs are in Articles 0.1–0.4.

4.1 Pillar P1 – Uniqueness and positivity of the ground state

The hypercube is connected, so Δ is irreducible. Adding the diagonal $\hat{V}_{m,n,k}$ does not change the off-diagonal pattern, so $H_{m,n,k}$ is irreducible. Consider the shifted matrix $M_{\text{shift}} = \alpha I - H_{m,n,k}$ with $\alpha = M^2 + 2M + 1$. Then M_{shift} has non-negative entries and is irreducible. By the Perron-Frobenius theorem, it has a simple largest eigenvalue ρ with a strictly positive eigenvector v . Then $H_{m,n,k}v = (\alpha - \rho)v$, and $\lambda_0 = \alpha - \rho$ is the smallest eigenvalue. Normalising gives the unique strictly positive ground state ψ_0 .

4.2 Pillar P2 – Constant spectral gap

The Kithima Bridge lemma implies that adding a diagonal non-negative potential cannot decrease the spectral gap. Since $\gamma(\Delta) = 2$, we have $\gamma(H_{m,n,k}) \geq 2$.

4.3 Pillar P3 – Logarithmic area law and MPS representation

The constant spectral gap implies exponential decay of correlations. By the Brandão–Horodecki theorem and a Hilbert curve ordering, the entanglement entropy satisfies $S(\rho_B) = O(\log M)$. Hence the maximum Schmidt rank is $e^{O(\log M)} = M^{O(1)}$. Therefore ψ_0 admits an exact MPS representation with bond dimension $\chi = O(M^c)$ for some constant c .

4.4 Pillar P4 – Deterministic extraction

The power iteration algorithm on MPS (Article I.5) converges to ψ_0 in $O(\log M)$ steps. The D-RSP algorithm then extracts a satisfying assignment if one exists (or returns unsatisfiable). The algorithm runs in time polynomial in M .

Theorem 4.1 (P4 for Fermat–Catalan Hamiltonian). *There exists a deterministic polynomial-time algorithm that, given the Hamiltonian $H_{m,n,k}$, decides the satisfiability of $\Phi_{m,n,k}$ and, if satisfiable, returns a satisfying assignment (i.e., a solution (a, b, c)).*

5 Application to each admissible triple

For each admissible exponent triple (m, n, k) (finitely many), we:

1. Construct the circuit $C_{m,n,k}$ (X.2) and the SAT instance $\Phi_{m,n,k}$.
2. Build the spectral Hamiltonian $H_{m,n,k}$ and compute its MPS representation (via power iteration).
3. Run the D-RSP algorithm to decide satisfiability.
4. If the solver returns a satisfying assignment, decode it to obtain (a, b, c) . This will be one of the known solutions (by the effective bound and the known classification). We then add a blocking clause to exclude that assignment and repeat to find all solutions. Since the total number of solutions is finite, the process terminates.

The combination of the effective bound and the spectral verification proves that the only solutions are the known ten.

6 Formalisation in Lean4

The Lean code imports the generic theorems from the kernel and instantiates them with the SAT instance for each triple.

Listing 1: SeriesX/FermatCatalanHamiltonian.lean (excerpt)

```
1 import X.FermatCatalanCircuit
2 import Tseitin
3 import SpectralLibrary
4 import KithimaBridge
5 import MPSAreaLaw
6 import PowerIteration
7
8 open SpectralLibrary
9
10 variable (m n k :      ) (B :      ) (hB : B      1)
11
12 def      : SATInstance := fermat_catalan_SAT m n k B hB
13 def M :      :=      .numVars
14 def Config := Fin M      Bool
15
16 def V (      : Config) :      :=
17   if satisfies      then 0 else M^2
18
19 def      : Matrix Config Config      := adjacencyMatrixOf (hypercube_graph
20   M)
```

```

21 def H_fc : Matrix Config Config      := diagonal V +
22
23 lemma H_irreducible : Irreducible H_fc :=
24   matrix_is_irreducible_of_adjacency_graph (hypercube_connected M)
25
26 theorem ground_state_unique_pos :
27   !      : Config      , (      ,      > 0)      (      ,      ^ 2
28     = 1)
29   (H_fc *      = ( _min H_fc)      ) :=
30   perron_frobenius H_fc
31
32 theorem spectral_gap_constant : spectral_gap H_fc      2 :=
33   kithima_bridge (diagonal V) (by simp [V, M])      (by simp) 1 (by
34     norm_num)
35
36 theorem area_law :      C,      B, entanglement_entropy (ground_state H_fc
37   ) B      C * Real.log M :=
38   area_law_for_hypercube (spectral_gap_constant)
39
40 theorem drsp_algorithm :      alg, polynomial_time alg
41   (      , satisfies      alg returns      ) :=
42   drsp_correct H_fc

```

All theorems are proved in the library; the file only instantiates them.

7 Conclusion

We have constructed the spectral Hamiltonian for each admissible exponent triple and verified the four pillars. The spectral SAT solver therefore applies. The next article (X.4) will execute the decision for all triples (finitely many) and conclude that no unknown solution exists, thereby proving the Fermat–Catalan conjecture.

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